

D 2 A

$$y = 1 + \ln x$$

$$- \ln x = 1 - y$$

$$\ln x = y - 1$$

$$x = e^{y-1}$$

D 2 B

$$y = 2 - \ln x$$

$$\ln x = 2 - y$$

$$x = e^{2-y}$$

D 2 C

$$y = \ln x - 3$$

$$- \ln x = -y - 3$$

$$\ln x = y + 3$$

$$x = e^{y+3}$$

D 3 A

$$a_n = \left(\frac{n-2}{n-1} \right)^n = (n-2) \cdot [(n-1)^n]^{-1} = e^{-2} \cdot e = e^{-1}$$

D 3 B

$$a_n = \left(\frac{n+2}{n+4} \right)^n = (n+2) \cdot [(n+4)^n]^{-1} = e^2 \cdot e^{-4} = e^{-2}$$

D 3 C

$$a_n = \left(\frac{n-3}{n-2} \right)^n = (n-3) \cdot [(n-2)^n]^{-1} = e^{-3} \cdot e^2 = e^{-1}$$

D 6 A

$$y = (\ln x)^2 - \ln x$$

$$y' = 2 \ln x \cdot \frac{1}{x} - \frac{1}{x} = \frac{2 \ln x - 1}{x}$$

$$\frac{2 \ln x - 1}{x} < 0 \Rightarrow \begin{cases} 2 \ln x - 1 < 0 \\ x > 0 \end{cases} \begin{cases} 2 \ln x < 1 \\ x > 0 \end{cases} \begin{cases} \ln x < 1/2 \\ x > 0 \end{cases} \begin{cases} x < \sqrt{e} \\ x > 0 \end{cases} \quad x \in (0, \sqrt{e})$$

D 6 B

$$y = 3 \ln x + (\ln x)^2$$

$$y' = 3/x + 2 \ln x \cdot \frac{1}{x} = \frac{3}{x} + \frac{2 \ln x}{x} = \frac{2 \ln x + 3}{x}$$

$$\frac{2 \ln x + 3}{x} < 0 \left\{ \begin{array}{l} 2 \ln x + 3 < 0 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} 2 \ln x < -3 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} \ln x < -3/2 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} x < e^{-3/2} \\ x > 0 \end{array} \right\} \quad x \in (0, e^{-3/2})$$

D 6 c

$$y = 2 \ln x + (\ln x)^2$$

$$y' = \frac{2}{x} + 2 \ln x \cdot \frac{1}{x} = \frac{2}{x} + \frac{2 \ln x}{x} = \frac{2 \ln x + 2}{x}$$

$$\frac{2 \ln x + 2}{x} < 0 \left\{ \begin{array}{l} 2 \ln x + 2 < 0 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} 2 \ln x < -2 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} \ln x < -1 \\ x > 0 \end{array} \right\} \left\{ \begin{array}{l} x < e^{-1} \\ x > 0 \end{array} \right\} \quad x \in (0, e^{-1})$$

D 9 A

$$\frac{n^2 + 2}{n^4 \ln n + 9} \sim \frac{1}{n^2 \ln n}$$

$$\frac{1}{n^2 \ln n} \text{ si comporta come } \frac{1}{n^2}$$

$$0 < 1/n^2 \ln n < 1/n^2$$

$1/n^2$ converge

la serie converge assolutamente

D 4 B

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^2 + 2}{n^4 \ln n + 9}$$

$$\frac{n^2 + 2}{n^4 \ln n + 9} \sim \frac{n^2}{n^4 \ln n} = \frac{1}{n^2 \ln n}$$